VENTURA COUNTY

Land Trust Expansion

Conservation and Restoration using Parcel Data

Client	Ventura Land Trust
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Course	ESRM 428 – Intermediate Geographic Information Systems
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Executive Summary

Ventura Land Trust (VLT) protects and restores open space, wildlife habitat, and natural resources in Ventura County. This project developed a parcel-based GIS prioritization framework to identify potential conservation and restoration opportunities using spatial relationships among legal parcels, existing VLT properties, rivers and streams, vegetation, wetlands, protected lands, and other project criteria. Five candidate sites were retained for final comparison. Site 1 and Site 2 received the highest reported suitability scores (100), followed by Site 3 (90), Site 4 (50), and Site 5 (50). Site 1 is recommended for the first stage of due-diligence review because, among the two top-scoring sites, its proximity to existing VLT land provides the stronger documented connectivity opportunity. The analysis is intended as decision support rather than an acquisition directive; ownership, cost, access, ecological condition, and long-term stewardship feasibility require verification before action.

Project Background

Ventura County contains coastal, chaparral, oak woodland, riparian, agricultural, urban, and foothill landscapes. Ventura Land Trust works to protect and restore open space and natural resources in this setting. Because potential conservation opportunities occur across many privately and publicly held parcels, GIS provides a practical framework for comparing spatial relationships consistently and narrowing the field to candidate parcels for additional review.

Management Question

Which parcels within the Ventura County study area provide the strongest conservation or restoration opportunities for Ventura Land Trust based on the spatial criteria implemented in this project?

Project Objectives

- Evaluate and prepare the client-provided and authoritative supplemental GIS datasets used in the analysis.
- Identify realistic candidate parcels using a transparent parcel-screening process.
- Quantify parcel relationships to existing VLT lands, hydrologic resources, habitat or vegetation, protected lands, and any other final criteria actually used.
- Develop a reproducible parcel-prioritization framework in ArcGIS Pro.
- Produce final maps, feature classes, a StoryMap, and an embedded Dashboard that communicate results to technical and general audiences.

Published StoryMap and Dashboard

Published StoryMap: VLT Storymap

Dashboard: Ventura Land Trust Purchase Recommendations

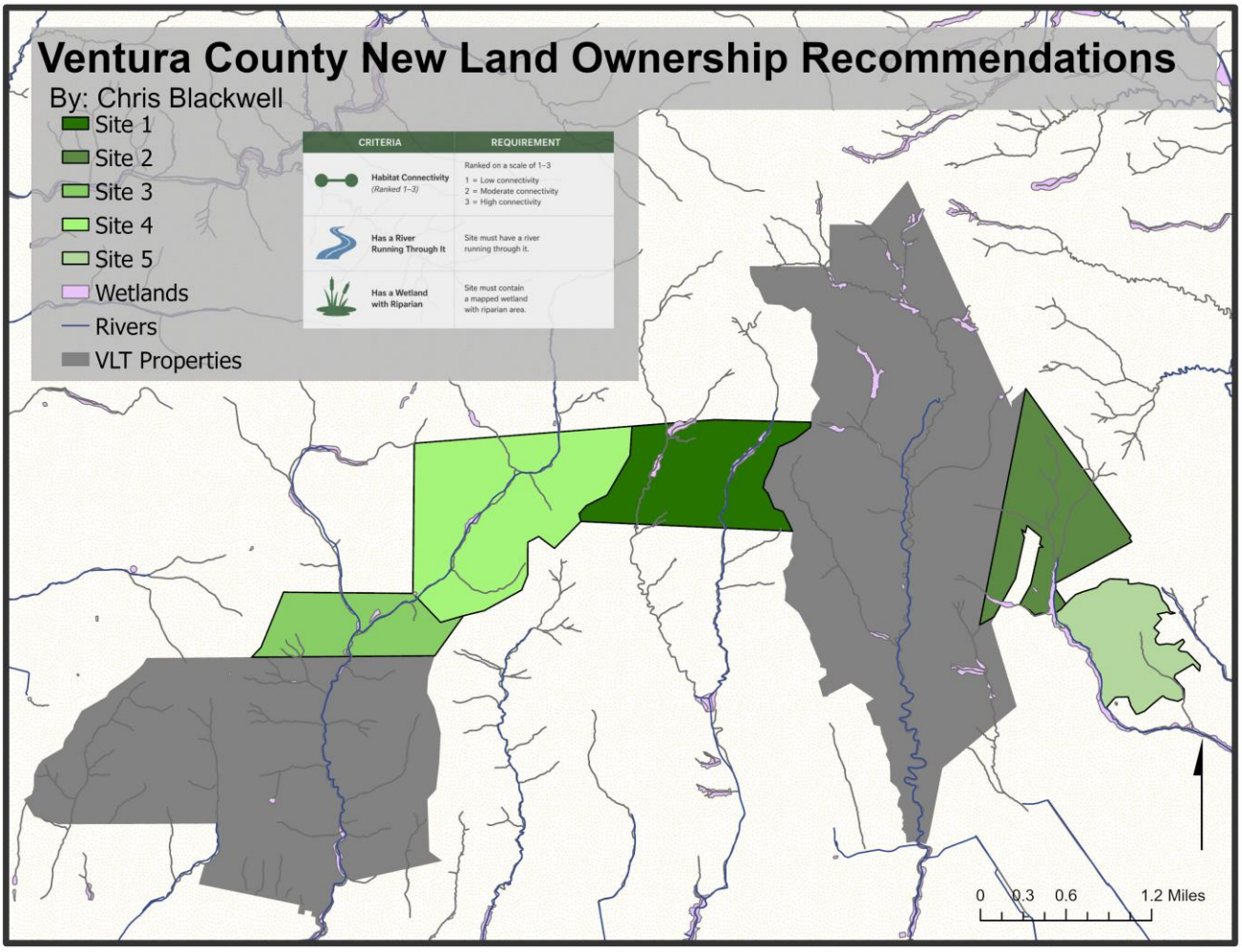


Figure 1. Final conservation/restoration parcel recommendations for the Ventura County study area.

Source Data

Source Data

Dataset Name	Data Source	Date Acquired	Purpose	Projection
Cities	VLT Data	7/18/2026	City boundaries for reference and mapping.	NAD83 StatePlane CA V
Municipalities	VLT Data	7/18/2026	Jurisdictional boundaries.	NAD83 StatePlane CA V
Natl_Forest_Boundary	VLT Data	7/18/2026	National Forest boundaries; excluded from suitable land.	NAD83 StatePlane CA V
Parcels	VLT Data	7/18/2026	Parcel geometry/attributes; primary analysis unit (polygon based).	WGS 1984
Rivers_Streams	VLT Data	7/18/2026	Lake Canyon River and tributaries; connectivity/intersection criterion.	NAD83 StatePlane CA V
Study_Area	VLT Data	7/18/2026	Extent of the study area.	NAD83 StatePlane CA V

Dataset Name	Data Source	Date Acquired	Purpose	Projection
VLT_AreaofInterest	VLT Data	7/18/2026	Specific area of land under study.	WGS 1984
VLT_Properties	VLT Data	7/18/2026	Existing VLT land; buffered at 100/200 ft for connectivity score.	WGS 1984

Table 1. Source data used in the Ventura County Land Trust Expansion analysis.

GIS Workflow and Methods

Data Preparation

- Import all source datasets into the project geodatabase.
- Use a common projected coordinate system: NAD 1983 StatePlane California V.
- Clip datasets to the Ventura County study area when necessary.
- Preserve original source data and create separate working feature classes for analysis.

Parcel Screening

- Use Ventura County parcels as the primary unit of analysis.
- Calculate or verify parcel acreage.
- Use Select By Attribute to remove parcels that do not meet the project screening requirements.
- Export the remaining parcels as the Candidate Parcels feature class.

VLT Property Connectivity

- Compare candidate parcels with existing Ventura Land Trust properties.
- State the exact proximity tool used in the completed ArcGIS Pro analysis and the distance or adjacency rule applied.
- Give greater priority to parcels that could expand or connect existing VLT-managed lands.

Protected-Land Connectivity

- Compare candidate parcels with existing protected lands.
- Identify parcels located adjacent to or near the existing conservation network.
- Exclude land that is already protected from being treated as a new acquisition opportunity.

Rivers and Streams Analysis

- Compare candidate parcels with the Rivers and Streams layer.
- State the exact hydrologic spatial-analysis tool used in the completed workflow and the intersection or proximity rule applied.
- Identify parcels that intersect or are close to waterways.
- Use this relationship to represent potential riparian and watershed conservation value.

Vegetation / Habitat Analysis

- Overlay candidate parcels with the vegetation or habitat dataset.
- Determine the vegetation types or amount of natural habitat occurring within each parcel.
- Assign a habitat value or score based on the vegetation criteria used in the project.

Wetlands

- Compare candidate parcels with mapped wetlands and/or floodplain areas.
- Identify parcels that overlap important water-resource areas.

- Incorporate this information into the conservation prioritization.

Parcel Prioritization

- Store each conservation criterion in a separate attribute field.
- Compare parcels using:
  - VLT connectivity
  - Protected-land connectivity
  - Rivers/stream proximity
  - Vegetation/habitat value
  - Wetland/floodplain value
- Combine the completed criteria into a final priority/suitability score.
- Rank parcels from higher to lower conservation priority.

Final Candidate Review

- Review the highest-priority parcels using aerial imagery and surrounding land uses.
- Check parcel configuration and relationships with VLT properties, waterways, habitat, and protected lands.
- Identify the strongest parcels for further conservation investigation.

Workflow Figure

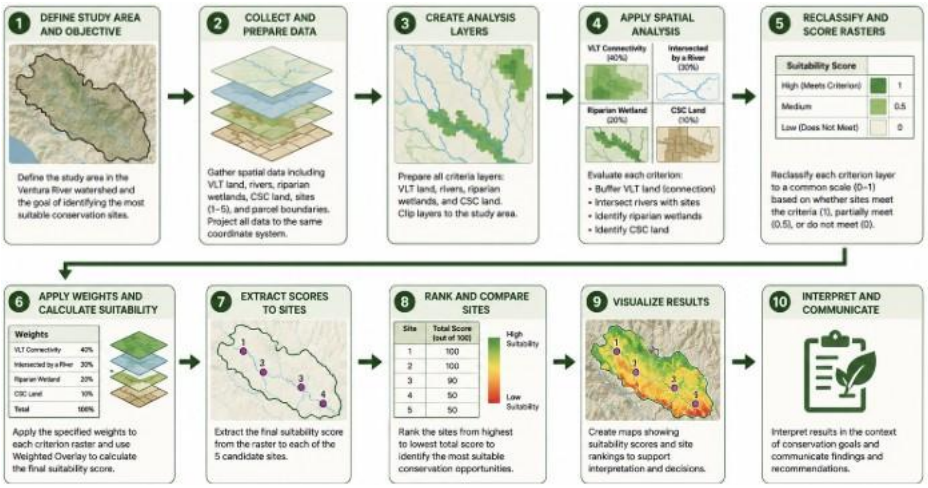


Figure 2. GIS workflow used to screen, evaluate, and prioritize candidate parcels for Ventura Land Trust. For final submission, the figure should show the actual ArcGIS Pro tools, inputs, intermediate outputs, scoring fields, and final candidate output used in this project.

Results

Result	Score
Site 1	100
Site 2	100
Site 3	90

Result	Score
Site 4	50
Site 5	50

Table 2. Results from the spatial analysis and suitability analysis.

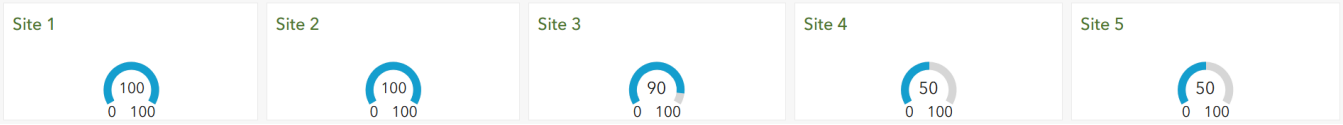


Figure 3. Overall analysis scores of each site.

Final Recommendation

Site 1 is recommended as the first site for VLT due-diligence review. Site 1 and Site 2 both received a reported suitability score of 100, so the total score alone does not distinguish them. Site 1 is ranked first because its documented proximity to existing VLT land provides the stronger habitat-connectivity and management-continuity rationale. This is a GIS-supported screening recommendation rather than an acquisition directive. Before action, VLT should verify ownership, legal access, easements or encumbrances, acquisition cost, biological condition, funding eligibility, and long-term stewardship feasibility.

Final Database

Final Database

General Dataset	File Name	Use / Description	Source / Derivation	Projection
Site 1	CB_VLT_FINAL.gdb	Site recommendation based on suitability scores and proximity to existing VLT land, where a new area of land might be purchased.	Acquired from Parcels data, from the VLT source data.	NAD83 StatePlane CA V
Site 2	CB_VLT_FINAL.gdb	Site recommendation based on suitability scores and proximity to existing VLT land, where a new area of land might be purchased.	Acquired from Parcels data, from the VLT source data.	NAD83 StatePlane CA V
Site 3	CB_VLT_FINAL.gdb	Site recommendation based on suitability scores and proximity to existing VLT land, where a new area of land might be purchased.	Acquired from Parcels data, from the VLT source data.	NAD83 StatePlane CA V
Site 4	CB_VLT_FINAL.gdb	Site recommendation based on suitability scores and proximity to existing VLT land, where a new area of land might be purchased.	Acquired from Parcels data, from the VLT source data.	NAD83 StatePlane CA V
Site 5	CB_VLT_FINAL.gdb	Site recommendation based on suitability scores and proximity to existing VLT land, where a new area of land might be purchased.	Acquired from Parcels data, from the VLT source data.	NAD83 StatePlane CA V
VC_Vegetation	CB_VLT_FINAL.gdb	Ability to tell which sections of land are CSC, Agriculture, Open Land, and CRS.	Acquired from Ventura County Open GIS data.	NAD83 StatePlane CA V
VC_Wetlands	CB_VLT_FINAL.gdb	Gives data on water use in the area and where water will be flowing, used to find riparian wetlands.	Acquired from Ventura County Open GIS data.	NAD83 StatePlane CA V

Table 3. Final feature classes produced by the analysis.

Data Dictionary

Field Name	Description
PARCEL_ID	Unique parcel identifier used for parcel-level reporting and identification.
LAND_USE	County land-use classification describing the current or designated use of the parcel.
CON_SCORE	Connectivity score based on the parcel's proximity to existing Ventura Land Trust properties.
VEG_TYPE	Dominant vegetation or habitat type identified within the parcel.
VEG_SCORE	Vegetation-priority score based on the conservation value of the vegetation or habitat occurring within the parcel.
RECOMMENDATION	Final planning category assigned to the parcel based on the GIS analysis and final review.

Table 4. Data dictionary providing input and clues to the data in the project.

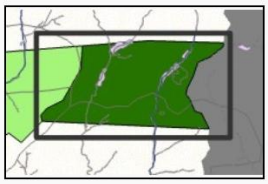
Updated Metadata

Site 1:

Item Description

Title: Site1

Thumbnail



Delete Update...

Tags

VLT

Summary (Purpose)

This is a separate layer that shows the overall size and has data on site 1, with the feature being created by clipping the feature class, then selecting parcels that are most related to the criteria for the project.

Site 2:

Item Description

Title: Site2

Thumbnail



Delete Update...

Tags

Tags

VLT

Summary (Purpose)

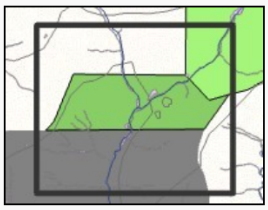
This is a separate layer that shows the overall size and has data on site 2, with the feature being created by clipping the feature class, then selecting parcels that are most related to the criteria for the project.

Description (Abstract)

Site 3:

Title: Site3

Thumbnail



Delete Update...

Tags

VLT

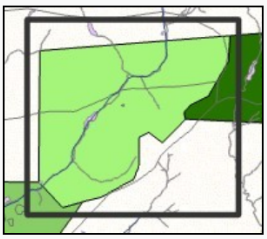
Summary (Purpose)

This is a separate layer that shows the overall size and has data on site 3, with the feature being created by clipping the feature class, then selecting parcels that are most related to the criteria for the project.

Site 4:

Title: Site4

Thumbnail



Delete Update...

Tags

VLT

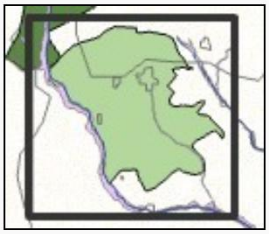
Summary (Purpose)

This is a separate layer that shows the overall size and has data on site 4, with the feature being created by clipping the feature class, then selecting parcels that are most related to the criteria for the project.

Site 5:

Title: Site5

Thumbnail



Delete Update...

Tags

VLT

Summary (Purpose)

This is a separate layer that shows the overall size and has data on site 5, with the feature being created by clipping the feature class, then selecting parcels that are most related to the criteria for the project.

VC_Vegetation

Title: Vegetation - Ventura County

Thumbnail



Delete Update...

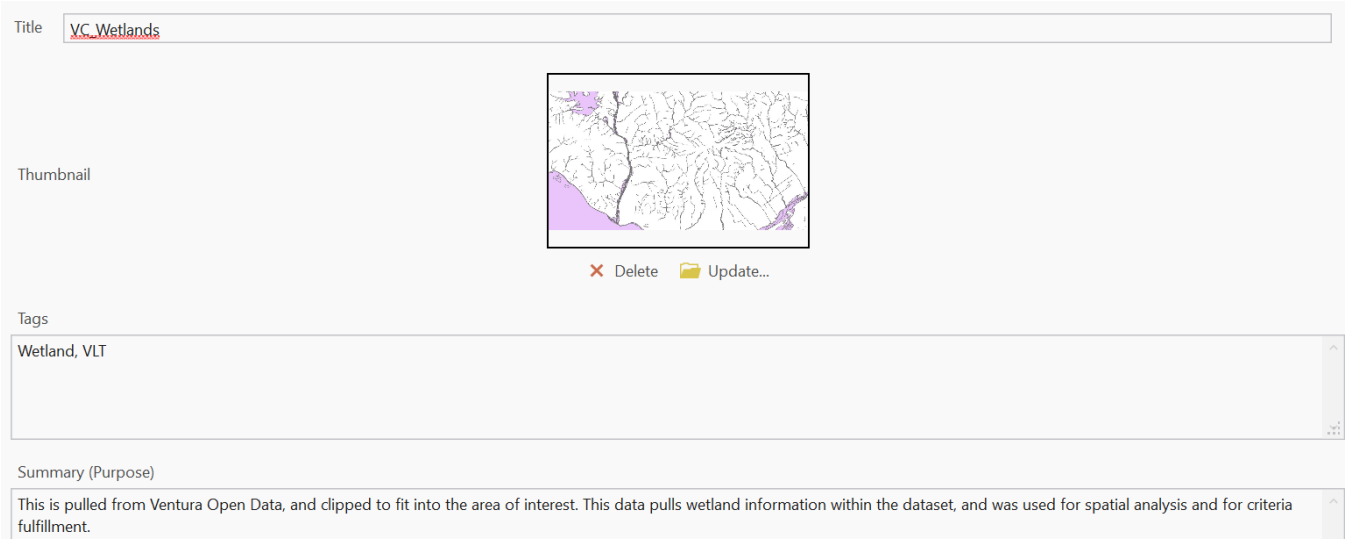
Tags

Ventura County, Vegetation

Summary (Purpose)

To create a GIS database of vegetation that covers the entire area of Ventura County. The database would be created from the most accurate sources available at the time. As additional or more accurate data are created, they will be incorporated into the vegetation database.

VC_Wetlands:



Limitations

- Spatial datasets may differ in date, scale, resolution, and positional accuracy; parcel-level recommendations should be field-verified.
- Any buffer or proximity thresholds are screening parameters unless supported by project-specific ecological literature.
- Ownership willingness, acquisition cost, access, easements, zoning, development rights, and long-term management costs are not fully represented by spatial suitability alone.

Discussion

This project demonstrates how GIS can support Ventura Land Trust's efforts to identify potential parcels for future conservation and land protection. Using parcels as the unit of analysis allows conservation characteristics to be connected directly to properties that could potentially be acquired, restored, protected through conservation easements, or investigated further. The analysis considers multiple factors, including proximity to existing VLT properties, protected lands, rivers and streams, vegetation or habitat, and parcel characteristics. Evaluating these factors together helps identify parcels that may provide multiple benefits, such as improving habitat connectivity while also protecting important watershed or riparian resources.

The results should be considered a decision support tool rather than a final acquisition plan. A parcel that performs well in the GIS analysis may still have limitations related to ownership, cost, access, development, legal restrictions, or long-term management. Environmental datasets also represent mapped conditions and may not completely reflect current conditions on the ground. For these reasons, the highest priority parcels should be treated as candidates for additional investigation. Field assessments, ownership research, biological surveys, and acquisition feasibility should be completed before VLT makes a final conservation decision.

Conclusion

The Ventura Land Trust conservation prioritization project developed a parcel based GIS approach for identifying potential conservation opportunities in Ventura County. By combining parcel boundaries with existing VLT properties, protected lands, waterways, vegetation or habitat information, and other conservation criteria, the project provides a systematic way to compare potential properties. Instead of relying on one environmental factor, the workflow identifies parcels where multiple conservation benefits overlap and provides a more transparent basis for prioritizing future conservation efforts.

Overall, the project provides VLT with a repeatable framework for narrowing a large study area into a smaller group of parcels worthy of further review. The final candidates are not automatic recommendations for acquisition, but they provide a starting point for field verification, landowner outreach, and additional ecological and financial evaluation.

References

- Ventura County GIS. Parcel, study area, municipal boundary, and rivers and streams datasets.
- Ventura Land Trust. Property holdings dataset supplied for the consulting project.
- California Protected Areas Database (CPAD), 2026 release.
- Esri. ArcGIS Pro, ArcGIS Online, ArcGIS StoryMaps, and ArcGIS Dashboards documentation.